

Week-1 Programming Practice — Lab Report

Compulsory, Medium & Advanced C Programs — Concepts, Explanation & Output Status

Level 1: Compulsory Programs

1. Check whether a given number is even or odd

Functions / Concepts Used:

Integer variable, scanf(), printf(), modulus operator (%), equality operator (==), if-else

Explanation:

- Reads an integer using scanf().
- Checks number % 2 == 0 using the modulus operator.
- if-else prints "Even" or "Odd" accordingly.

✓ Program successfully executed.

2. Check whether a given number is positive, negative, or zero

Functions / Concepts Used:

Variables, relational operators (>, <, ==), if-else-if ladder

Explanation:

- Reads a number into a variable.
- Compares it with 0 using >, <, == in an if-else-if ladder.
- Prints Positive, Negative, or Zero based on the match.

✓ Program successfully executed.

3. Check whether a character is uppercase, lowercase, digit, or special character

Functions / Concepts Used:

char data type, character input, relational operators, logical AND (&&), else-if ladder

Explanation:

- Reads a character using the char data type.
- Uses relational operators with && to check A-Z, a-z, 0-9 ranges.
- else-if ladder prints the matching category.

✓ Program successfully executed.

4. Check whether a given year is a leap year

Functions / Concepts Used:

Integer variable, modulus operator (%), relational and logical operators, nested conditions

Explanation:

- Reads the year as an integer.
- Checks divisibility by 4, 100, 400 using % and nested conditions.
- Prints Leap Year or Not a Leap Year based on the rule.

✓ Program successfully executed.

5. Print the memory allocation for all data types in C

Functions / Concepts Used:

Data types, printf(), format specifiers, sizeof operator

Explanation:

- Declares variables of int, char, float, double, etc.
- Uses sizeof() to get bytes occupied by each type.
- printf() with format specifiers displays each size.

✓ Program successfully executed.

Level 2: Medium Programs

1. Menu-based arithmetic program using switch

Functions / Concepts Used: